Lesson 16: Normed Spaces and Banach Spaces

AI/ML Mathematics

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1 Normed Spaces

1.1 Definition and Basic Properties

Definition 1.1 (Normed Space)**.** A *normed space* (or *normed vector space*) is a pair $(X, \|\cdot\|)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\|\cdot\|\colon X \to [0,\infty)$ is a *norm*, satisfying for all $x, y \in X$ and scalars α:

(N1) $\|x\| = 0$ if and only if $x = 0$. (positive definiteness)
(N2) $\|\alpha x\| = |\alpha|\,\|x\|$. (absolute homogeneity)
(N3) $\|x + y\| \le \|x\| + \|y\|$. (triangle inequality)

Remark 1.2. Non-negativity follows from the axioms: $0 = \|x + (-x)\| \le \|x\| + \|-x\| = 2\|x\|$ by (N2) with $\alpha = -1$, so $\|x\| \ge 0$. Also, the reverse triangle inequality $|\|x\| - \|y\|| \le \|x - y\|$ follows by applying (N3) to $x = (x - y) + y$ and $y = (y - x) + x$.

Proposition 1.3 (Every Norm Induces a Metric)**.** *If $(X, \|\cdot\|)$ is a normed space, then $d(x, y) = \|x - y\|$ defines a metric on X. This metric is* translation-invariant *($d(x + z, y + z) = d(x, y)$) and* homogeneous *($d(\alpha x, \alpha y) = |\alpha|\, d(x, y)$).*

Proof. **(M1):** $d(x, y) = \|x - y\| = 0 \iff x - y = 0 \iff x = y$ by (N1). **(M2):** $d(x, y) = \|x - y\| = \|-(y - x)\| = |-1|\|y - x\| = d(y, x)$ by (N2). **(M3):** $d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \le \|x - y\| + \|y - z\| = d(x, y) + d(y, z)$ by (N3). Translation invariance: $d(x+z, y+z) = \|(x+z)-(y+z)\| = \|x-y\| = d(x, y)$. Homogeneity: $d(\alpha x, \alpha y) = \|\alpha(x-y)\| = |\alpha|\|x - y\| = |\alpha|\, d(x, y)$ by (N2). □

1.2 Important Examples of Normed Spaces

Example 1.4 ($\mathbb{R}^n$ with p-Norms)**.** For $1 \le p \le \infty$, $(\mathbb{R}^n, \|\cdot\|_p)$ is a normed space, where $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ for $p < \infty$ and $\|x\|_\infty = \max_i |x_i|$. Axioms (N1) and (N2) are immediate. The triangle inequality for $1 < p < \infty$ is Minkowski's inequality (Proposition 2.3); for $p = \infty$: $\|x + y\|_\infty = \max_i |x_i + y_i| \le \max_i |x_i| + \max_i |y_i|$.

Example 1.5 ($C[a, b]$ with Supremum and L^1 Norms)**.** $C[a, b]$ with $\|f\|_\infty = \max_{t\in[a,b]} |f(t)|$ is a normed space (the max exists by the extreme value theorem). Alternatively, $\|f\|_1 = \int_a^b |f(t)|\, dt$ also defines a norm on $C[a, b]$, but $C[a, b]$ is *not* complete under $\|\cdot\|_1$.

Example 1.6 (Sequence Spaces ℓ^p)**.** For $1 \le p \le \infty$, $(\ell^p, \|\cdot\|_p)$ is a normed space by Minkowski's inequality (extended to infinite sums), where $\ell^p = \{x = (x_n) : \|x\|_p < \infty\}$.

2 The L^p Inequality Chain

To verify that $\|\cdot\|_p$ satisfies the triangle inequality for $1 < p < \infty$, we establish the chain Young $\Rightarrow$ Hölder $\Rightarrow$ Minkowski.

2.1 Young's Inequality

Lemma 2.1 (Young's Inequality)**.** *For $a, b \ge 0$ and $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$:*

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}.$$

Equality holds if and only if $a^p = b^q$.

Proof. If $a = 0$ or $b = 0$, the inequality is trivial (both sides are non-negative and the left side is zero). Assume $a, b > 0$.

The exponential function $t \mapsto e^t$ is convex. By Jensen's inequality (or equivalently, by the definition of convexity applied to two points):

$$e^{\lambda s+(1-\lambda)t} \leq \lambda\, e^s + (1-\lambda)\, e^t \quad \text{for all } s, t \in \mathbb{R},\ \lambda \in [0,1].$$

Set $\lambda = 1/p$, so $1 - \lambda = 1/q$. Set $s = p \ln a$ and $t = q \ln b$. Then:

$$\begin{aligned} ab &= e^{\ln a + \ln b} = e^{(1/p)(p \ln a)+(1/q)(q \ln b)} \\ &\leq \frac{1}{p}\, e^{p \ln a} + \frac{1}{q}\, e^{q \ln b} = \frac{a^p}{p} + \frac{b^q}{q}. \end{aligned}$$

For the equality condition: equality holds in Jensen's inequality if and only if $s = t$, i.e., $p \ln a = q \ln b$, i.e., $a^p = b^q$. □

2.2 Hölder's Inequality

Proposition 2.2 (Hölder's Inequality)**.** *Let $1 < p < \infty$ and $q = p/(p-1)$ (so $1/p + 1/q = 1$). For $x = (x_i)$, $y = (y_i)$ in $\mathbb{R}^n$ (or sequences in ℓ^p, ℓ^q):*

$$\sum_i |x_i y_i| \leq \|x\|_p\, \|y\|_q.$$

Equality holds if and only if there exist constants $\alpha, \beta \geq 0$ (not both zero) with $\alpha|x_i|^p = \beta|y_i|^q$ for all i.

Proof. If $\|x\|_p = 0$ then $x = 0$ and both sides are zero. Similarly for $\|y\|_q = 0$. Assume $\|x\|_p > 0$ and $\|y\|_q > 0$.

Define the normalized vectors $a_i = |x_i|/\|x\|_p$ and $b_i = |y_i|/\|y\|_q$. Apply Young's inequality (Lemma 2.1) to each pair (a_i, b_i):

$$a_i b_i \leq \frac{a_i^p}{p} + \frac{b_i^q}{q}.$$

Sum over i:

$$\sum_i a_i b_i \leq \frac{1}{p} \sum_i \frac{|x_i|^p}{\|x\|_p^p} + \frac{1}{q} \sum_i \frac{|y_i|^q}{\|y\|_q^q} = \frac{1}{p} \cdot 1 + \frac{1}{q} \cdot 1 = 1.$$

That is,

$$\sum_i \frac{|x_i|}{\|x\|_p} \cdot \frac{|y_i|}{\|y\|_q} \leq 1.$$

Multiplying both sides by $\|x\|_p \|y\|_q$ gives the result. □

2.3 Minkowski's Inequality

Proposition 2.3 (Minkowski's Inequality)**.** *For $1 \leq p < \infty$ and $x, y \in \mathbb{R}^n$ (or ℓ^p):*

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p.$$

Proof. **Case $p = 1$:** $\|x + y\|_1 = \sum_i |x_i + y_i| \leq \sum_i (|x_i| + |y_i|) = \|x\|_1 + \|y\|_1$ by the ordinary triangle inequality for real numbers.

Case $1 < p < \infty$: Let $q = p/(p-1)$, so $1/p + 1/q = 1$ and $(p-1)q = p$. If $\|x + y\|_p = 0$, the inequality is trivial. Assume $\|x + y\|_p > 0$. We compute:

$$\begin{aligned} \|x + y\|_p^p &= \sum_i |x_i + y_i|^p = \sum_i |x_i + y_i|^{p-1} |x_i + y_i| \\ &\leq \sum_i |x_i + y_i|^{p-1} |x_i| + \sum_i |x_i + y_i|^{p-1} |y_i|. \end{aligned}$$

Apply Hölder's inequality (Proposition 2.2) to each sum. The sequence $(|x_i + y_i|^{p-1})_i$ has ℓ^q norm:

$$\Big(\sum_i \big(|x_i + y_i|^{p-1}\big)^q\Big)^{1/q} = \Big(\sum_i |x_i + y_i|^{(p-1)q}\Big)^{1/q} = \Big(\sum_i |x_i + y_i|^p\Big)^{1/q} = \|x + y\|_p^{p/q}.$$

Therefore:

$$\sum_i |x_i + y_i|^{p-1}|x_i| \le \|x + y\|_p^{p/q}\, \|x\|_p,$$
$$\sum_i |x_i + y_i|^{p-1}|y_i| \le \|x + y\|_p^{p/q}\, \|y\|_p.$$

Combining:

$$\|x + y\|_p^p \le \|x + y\|_p^{p/q}\big(\|x\|_p + \|y\|_p\big).$$

Dividing both sides by $\|x+y\|_p^{p/q}$ (which is positive) and using $p - p/q = p(1-1/q) = p \cdot 1/p = 1$:

$$\|x + y\|_p^{p-p/q} = \|x + y\|_p \le \|x\|_p + \|y\|_p. \qquad \square$$

The L^p Inequality Chain

Young's inequality $\Rightarrow$ Hölder's inequality $\Rightarrow$ Minkowski's inequality. This chain establishes that $\|\cdot\|_p$ is a valid norm for $1 \le p \le \infty$, on both $\mathbb{R}^n$ and ℓ^p. Every step in the chain has a clear logical dependency: Young provides the pointwise bound, Hölder aggregates it into a sum, and Minkowski converts it into the triangle inequality.

3 Equivalence of Norms in Finite Dimensions

Definition 3.1 (Equivalent Norms)**.** Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a vector space X are *equivalent* if there exist constants $c, C > 0$ such that

$$c\,\|x\|_a \le \|x\|_b \le C\,\|x\|_a \quad \text{for all } x \in X.$$

Remark 3.2. Equivalent norms define the same topology: they determine the same open sets, convergent sequences, Cauchy sequences, and continuous functions. In particular, a space is complete under one norm if and only if it is complete under any equivalent norm. Equivalence of norms is an equivalence relation (reflexivity, symmetry, and transitivity are straightforward).

Theorem 3.3 (All Norms on $\mathbb{R}^n$ Are Equivalent)**.** *On $\mathbb{R}^n$, every two norms are equivalent.*

Proof. It suffices to show that any norm $\|\cdot\|$ on $\mathbb{R}^n$ is equivalent to the fixed norm $\|\cdot\|_\infty$, since equivalence is transitive.

Step 1: Upper bound. Let $e_1, \dots, e_n$ be the standard basis vectors. For $x = \sum_{i=1}^n x_i e_i$:

$$\|x\| = \Big\|\sum_{i=1}^n x_i e_i\Big\| \le \sum_{i=1}^n |x_i|\,\|e_i\| \le \Big(\max_{1\le i\le n} |x_i|\Big) \sum_{i=1}^n \|e_i\| = C\,\|x\|_\infty,$$

where $C = \sum_{i=1}^n \|e_i\| > 0$.

Step 2: Continuity. The map $f\colon (\mathbb{R}^n, \|\cdot\|_\infty) \to \mathbb{R}$ defined by $f(x) = \|x\|$ is Lipschitz continuous:

$$|f(x) - f(y)| = \big|\|x\| - \|y\|\big| \le \|x - y\| \le C\,\|x - y\|_\infty,$$

where the first inequality is the reverse triangle inequality for $\|\cdot\|$ and the second is Step 1.

Step 3: Lower bound via compactness. The unit sphere $S = \{x \in \mathbb{R}^n : \|x\|_\infty = 1\}$ is closed and bounded in $(\mathbb{R}^n, \|\cdot\|_\infty)$, hence compact by the Heine–Borel theorem. The continuous function $f(x) = \|x\|$ attains its minimum on S: let $c = \min_{x\in S} f(x) = \min_{x\in S} \|x\|$.

We claim $c > 0$. If $x \in S$, then $\|x\|_\infty = 1$, so $x \neq 0$, and therefore $\|x\| > 0$ by (N1). Since f is continuous on the compact set S and strictly positive on S, its minimum c is strictly positive.

For any $x \neq 0$, the vector $x/\|x\|_\infty \in S$, so:

$$\left\| \frac{x}{\|x\|_\infty} \right\| \geq c,$$

which by homogeneity gives $\|x\| \geq c\,\|x\|_\infty$. This inequality holds trivially for $x = 0$ as well.

Conclusion. Combining Steps 1 and 3: $c\,\|x\|_\infty \leq \|x\| \leq C\,\|x\|_\infty$ for all $x \in \mathbb{R}^n$. □

Finite-Dimensional Norm Equivalence

On a finite-dimensional vector space, all norms are equivalent. This means convergence, completeness, compactness, and continuity are *norm-independent* in $\mathbb{R}^n$. This fails dramatically in infinite dimensions.

Corollary 3.4 (Finite-Dimensional Normed Spaces Are Complete)**.** *Every finite-dimensional normed space is a Banach space.*

Proof. Let $(X, \|\cdot\|)$ be a finite-dimensional normed space of dimension n. Fix a basis $e_1, \ldots, e_n$ for X and consider the coordinate isomorphism $\Phi\colon \mathbb{R}^n \to X$ defined by $\Phi(\alpha_1, \ldots, \alpha_n) = \sum_i \alpha_i e_i$. The map Φ is a linear bijection, and $\|\cdot\|_X := \|\cdot\| \circ \Phi$ defines a norm on $\mathbb{R}^n$.

By Theorem 3.3, $\|\cdot\|_X$ is equivalent to $\|\cdot\|_\infty$ on $\mathbb{R}^n$. Since $(\mathbb{R}^n, \|\cdot\|_\infty)$ is complete (coordinatewise limits of Cauchy sequences converge), and equivalent norms share the same Cauchy sequences and convergent sequences (Remark 3.2), $(\mathbb{R}^n, \|\cdot\|_X)$ is complete. Since Φ is an isometric isomorphism between $(\mathbb{R}^n, \|\cdot\|_X)$ and $(X, \|\cdot\|)$, the space $(X, \|\cdot\|)$ is complete. □

Norms on Infinite-Dimensional Spaces Are Not Equivalent

On $C[0,1]$, the norms $\|f\|_\infty = \max|f|$ and $\|f\|_1 = \int_0^1 |f|$ are *not* equivalent. Consider $f_n(t) = t^n$: then $\|f_n\|_\infty = 1$ for all n, but $\|f_n\|_1 = 1/(n+1) \to 0$. If the norms were equivalent, we would have $c\,\|f_n\|_\infty \leq \|f_n\|_1$ for some $c > 0$, giving $c \leq 1/(n+1) \to 0$, a contradiction. Consequently, $C[0,1]$ is complete under $\|\cdot\|_\infty$ but *not* under $\|\cdot\|_1$.

Norm Choice in RL

Value iteration contracts in $\|\cdot\|_\infty$ with factor γ, but need not contract in $\|\cdot\|_2$. For finite state spaces all norms are equivalent, so convergence in one implies convergence in all. For infinite state spaces with function approximation, non-equivalence of norms is at the root of many difficulties in approximate dynamic programming.

4 Banach Spaces

Definition 4.1 (Banach Space)**.** A *Banach space* is a complete normed space, i.e., a normed space $(X, \|\cdot\|)$ in which every Cauchy sequence converges (with respect to the metric $d(x,y) = \|x - y\|$).

4.1 Completeness of ℓ^p

Theorem 4.2 (ℓ^p Is a Banach Space)**.** *For $1 \le p \le \infty$, the space $(\ell^p, \|\cdot\|_p)$ is a Banach space.*

Proof. We give the proof for $1 \le p < \infty$; the case $p = \infty$ is analogous (replacing sums with suprema).

Let $(x^{(k)})_{k=1}^{\infty}$ be a Cauchy sequence in ℓ^p, where $x^{(k)} = (x_1^{(k)}, x_2^{(k)}, \ldots)$.

Step 1: Identify the candidate limit. For each fixed index $i \in \mathbb{N}$,

$$|x_i^{(k)} - x_i^{(m)}| \le \Big(\sum_{j=1}^{\infty} |x_j^{(k)} - x_j^{(m)}|^p\Big)^{1/p} = \|x^{(k)} - x^{(m)}\|_p \to 0 \quad \text{as } k, m \to \infty.$$

So $(x_i^{(k)})_{k=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}$ for each i, and by completeness of $\mathbb{R}$, it converges. Define $x_i = \lim_{k\to\infty} x_i^{(k)}$ and set $x = (x_1, x_2, \ldots)$.

Step 2: Show $x \in \ell^p$ and $x^{(k)} \to x$. Fix $\varepsilon > 0$ and choose $K \in \mathbb{N}$ such that $\|x^{(k)} - x^{(m)}\|_p < \varepsilon$ for all $k, m \ge K$. For any finite $N \in \mathbb{N}$ and $k, m \ge K$:

$$\Big(\sum_{i=1}^{N} |x_i^{(k)} - x_i^{(m)}|^p\Big)^{1/p} \le \|x^{(k)} - x^{(m)}\|_p < \varepsilon.$$

Fix $k \ge K$ and let $m \to \infty$. By continuity of $|\cdot|^p$ and of finite sums:

$$\Big(\sum_{i=1}^{N} |x_i^{(k)} - x_i|^p\Big)^{1/p} \le \varepsilon.$$

Since this holds for every N, taking $N \to \infty$:

$$\|x^{(k)} - x\|_p = \Big(\sum_{i=1}^{\infty} |x_i^{(k)} - x_i|^p\Big)^{1/p} \le \varepsilon.$$

In particular, $x^{(k)} - x \in \ell^p$, and by the triangle inequality $\|x\|_p \le \|x - x^{(k)}\|_p + \|x^{(k)}\|_p \le \varepsilon + \|x^{(K)}\|_p < \infty$, so $x \in \ell^p$.

Step 3: Convergence. From Step 2, $\|x^{(k)} - x\|_p \le \varepsilon$ for all $k \ge K$. Since ε was arbitrary, $x^{(k)} \to x$ in ℓ^p. □

4.2 Completeness of $C[a, b]$

Theorem 4.3 ($C[a, b]$ Is Banach Under the Sup Norm)**.** *$(C[a, b], \|\cdot\|_\infty)$ is a Banach space.*

Proof. Let $(f_n)_{n=1}^{\infty}$ be a Cauchy sequence in $(C[a, b], \|\cdot\|_\infty)$.

Step 1: Pointwise convergence. For each $t \in [a, b]$, $|f_n(t) - f_m(t)| \le \|f_n - f_m\|_\infty \to 0$, so $(f_n(t))_n$ is Cauchy in $\mathbb{R}$, hence convergent. Define $f(t) = \lim_{n\to\infty} f_n(t)$.

Step 2: Uniform convergence. Given $\varepsilon > 0$, choose N such that $\|f_n - f_m\|_\infty < \varepsilon$ for all $n, m \ge N$. For each $t \in [a, b]$ and all $n \ge N$: $|f_n(t) - f_m(t)| < \varepsilon$ for all $m \ge N$. Letting $m \to \infty$: $|f_n(t) - f(t)| \le \varepsilon$. Since this holds for *all* $t \in [a, b]$, we have $\|f_n - f\|_\infty \le \varepsilon$. Hence $f_n \to f$ uniformly on $[a, b]$.

Step 3: The limit is continuous. The uniform limit of continuous functions is continuous. Fix $t_0 \in [a, b]$ and $\varepsilon > 0$. Choose n with $\|f_n - f\|_\infty < \varepsilon/3$, then $\delta > 0$ with $|f_n(t) - f_n(t_0)| < \varepsilon/3$ for $|t - t_0| < \delta$ (by continuity of f_n). For $|t - t_0| < \delta$:

$$|f(t) - f(t_0)| \le |f(t) - f_n(t)| + |f_n(t) - f_n(t_0)| + |f_n(t_0) - f(t_0)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

So $f \in C[a, b]$ and $f_n \to f$ in $\|\cdot\|_\infty$. □

Example 4.4 ($C[a,b]$ Is Not Complete Under $\|\cdot\|_1$)**.** The space $(C[0,1], \|\cdot\|_1)$ is *not* complete. Define $f_n\colon [0,1] \to \mathbb{R}$ by

$$f_n(t) = \begin{cases} 0 & \text{if } 0 \le t \le \frac{1}{2}, \\ n(t-\frac{1}{2}) & \text{if } \frac{1}{2} < t \le \frac{1}{2} + \frac{1}{n}, \\ 1 & \text{if } \frac{1}{2} + \frac{1}{n} < t \le 1. \end{cases}$$

Each f_n is continuous, and for $m > n$: $\|f_n - f_m\|_1 = \int_0^1 |f_n - f_m| \le 1/n \to 0$, so (f_n) is Cauchy. The pointwise limit is the discontinuous step function $\mathbf{1}_{(1/2,1]}$, which is not in $C[0,1]$. Moreover, no continuous function can be the L^1-limit: if $g \in C[0,1]$ with $\|f_n - g\|_1 \to 0$, then $g = \mathbf{1}_{(1/2,1]}$ almost everywhere, contradicting continuity.

5 Series in Banach Spaces

Definition 5.1 (Absolute Convergence)**.** A series $\sum_{n=1}^{\infty} x_n$ in a normed space $(X, \|\cdot\|)$ is *absolutely convergent* if $\sum_{n=1}^{\infty} \|x_n\| < \infty$. It is *convergent* if the partial sums $S_N = \sum_{n=1}^{N} x_n$ converge in norm to some $S \in X$.

Remark 5.2. In $\mathbb{R}$ (or $\mathbb{C}$), absolute convergence implies convergence. This is essentially a restatement of completeness. In a general normed space, absolute convergence need not imply convergence – the space must be complete.

Theorem 5.3 (Characterization of Banach Spaces via Series)**.** *A normed space $(X, \|\cdot\|)$ is a Banach space if and only if every absolutely convergent series in X converges.*

Proof. ($\Rightarrow$) Suppose X is Banach and $\sum_{n=1}^{\infty} \|x_n\| < \infty$. We show that the partial sums $S_N = \sum_{n=1}^{N} x_n$ form a Cauchy sequence. For $M > N$:

$$\|S_M - S_N\| = \Big\| \sum_{n=N+1}^{M} x_n \Big\| \le \sum_{n=N+1}^{M} \|x_n\| \le \sum_{n=N+1}^{\infty} \|x_n\| \to 0$$

as $N \to \infty$ (tail of a convergent real series). Since X is complete, the Cauchy sequence (S_N) converges.

($\Leftarrow$) Suppose every absolutely convergent series converges. Let $(y_n)_{n=1}^{\infty}$ be a Cauchy sequence in X. We must show that (y_n) converges.

Step 1: Extract a fast subsequence. Since (y_n) is Cauchy, for each $k \ge 1$ there exists n_k such that $\|y_m - y_\ell\| < 2^{-k}$ for all $m, \ell \ge n_k$. By choosing $n_1 < n_2 < n_3 < \cdots$ (taking n_k large enough at each step), we obtain a subsequence $(y_{n_k})_{k=1}^{\infty}$ with

$$\|y_{n_{k+1}} - y_{n_k}\| < 2^{-k} \quad \text{for all } k \ge 1.$$

Step 2: Write the subsequence as a telescoping series. Define $z_1 = y_{n_1}$ and $z_k = y_{n_k} - y_{n_{k-1}}$ for $k \ge 2$. Then the partial sums satisfy $\sum_{k=1}^{K} z_k = y_{n_K}$, and:

$$\sum_{k=1}^{\infty} \|z_k\| = \|y_{n_1}\| + \sum_{k=2}^{\infty} \|y_{n_k} - y_{n_{k-1}}\| \le \|y_{n_1}\| + \sum_{k=1}^{\infty} 2^{-k} = \|y_{n_1}\| + 1 < \infty.$$

So $\sum z_k$ converges absolutely, hence converges by hypothesis: there exists $y \in X$ with $y_{n_K} = \sum_{k=1}^{K} z_k \to y$.

Step 3: The full sequence converges. Given $\varepsilon > 0$, choose N_0 with $\|y_m - y_\ell\| < \varepsilon/2$ for $m, \ell \ge N_0$, and K_0 with $\|y_{n_K} - y\| < \varepsilon/2$ for $K \ge K_0$. For $n \ge N_0$, pick $K \ge K_0$ with $n_K \ge N_0$. Then:

$$\|y_n - y\| \le \|y_n - y_{n_K}\| + \|y_{n_K} - y\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $y_n \to y$, and X is complete. $\square$

Banach Space Characterization

A normed space is Banach if and only if every absolutely convergent series converges. This characterization is often the most convenient way to verify completeness, because one constructs a convergent series rather than chasing arbitrary Cauchy sequences.

Why Completeness Matters for RL

The Banach fixed-point theorem guarantees that a contraction on a complete metric space has a unique fixed point. The Bellman operator is a γ-contraction on the Banach space of bounded functions. Without completeness, the value iterates might "converge" to a limit outside the space. This is why we work in Banach spaces, not merely normed spaces.

6 Schauder Basis

In finite-dimensional spaces, every vector has a unique representation as a finite linear combination of basis vectors. In infinite-dimensional Banach spaces, we replace finite sums with convergent series.

Definition 6.1 (Schauder Basis)**.** A sequence $(e_n)_{n=1}^{\infty}$ in a Banach space X is a *Schauder basis* if every $x \in X$ has a unique representation

$$x = \sum_{n=1}^{\infty} \alpha_n e_n,$$

where the series converges in norm and the coefficients $\alpha_n \in \mathbb{R}$ are uniquely determined by x.

Example 6.2 (Schauder Basis for ℓ^p)**.** The standard unit vectors $e_n = (0, \ldots, 0, 1, 0, \ldots)$ (with 1 in position n) form a Schauder basis for ℓ^p when $1 \le p < \infty$. Indeed, every $x = (x_1, x_2, \ldots) \in \ell^p$ satisfies:

$$\Big\| x - \sum_{n=1}^{N} x_n e_n \Big\|_p = \Big(\sum_{n=N+1}^{\infty} |x_n|^p \Big)^{1/p} \to 0 \quad \text{as } N \to \infty,$$

since $\sum |x_n|^p < \infty$. The coefficients are unique because if $\sum \alpha_n e_n = \sum \beta_n e_n$, evaluating at coordinate k gives $\alpha_k = \beta_k$.

Remark 6.3. The standard unit vectors do *not* form a Schauder basis for ℓ^∞: for $x = (1, 1, 1, \ldots)$, $\|x - \sum_{n=1}^{N} e_n\|_\infty = 1$ for all N. More generally, ℓ^∞ has no Schauder basis because it is not separable.

7 Exercises

Exercise 7.1 (Parallelogram Law)**.** Prove that in any inner product space (a normed space whose norm arises from an inner product $\|x\|^2 = \langle x, x \rangle$), the following identity holds:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \text{for all } x, y.$$

Show that $\|\cdot\|_p$ on $\mathbb{R}^n$ (with $n \ge 2$) satisfies the parallelogram law if and only if $p = 2$.

Hint: For the second part, test with $x = e_1$ and $y = e_2$.

Exercise 7.2 (Completeness of ℓ^∞)**.** Prove directly (without using Theorem 5.3) that $(\ell^\infty, \|\cdot\|_\infty)$ is a Banach space.

Hint: Let $(x^{(k)})$ be Cauchy in ℓ^∞. Show that coordinatewise limits exist, the limit sequence is bounded, and convergence holds in $\|\cdot\|_\infty$.

Exercise 7.3 (Norm Equivalence on $\mathbb{R}^2$)**.** On $\mathbb{R}^2$, find the best (sharpest) constants c and C such that $c\,\|x\|_2 \le \|x\|_1 \le C\,\|x\|_2$ for all $x \in \mathbb{R}^2$.

Hint: For the upper bound, use Cauchy–Schwarz. For the lower bound, find the minimum of $\|x\|_1/\|x\|_2$ on the unit circle.

Exercise 7.4 (Banach Characterization Application)**.** Use Theorem 5.3 to prove that the space $c_0 = \{x \in \ell^\infty : \lim_{n\to\infty} x_n = 0\}$ with the sup norm $\|\cdot\|_\infty$ is a Banach space.

Hint: Let $\sum_{k=1}^{\infty} x^{(k)}$ be absolutely convergent in c_0. First show the series converges in ℓ^∞ (which is Banach). Then show the limit belongs to c_0, i.e., it is a sequence converging to zero.

Exercise 7.5 (Hölder's Inequality: Equality Case)**.** Let $1 < p < \infty$ with $1/p + 1/q = 1$, and let $x \in \ell^p$, $y \in \ell^q$ with $\|x\|_p = \|y\|_q = 1$. Prove that $\sum_{i=1}^{\infty} |x_i y_i| = 1$ if and only if $|x_i|^p = |y_i|^q$ for all i.

Hint: Trace back through the proof of Proposition 2.2 and use the equality condition in Young's inequality (Lemma 2.1).

References